

1. Introduction

Arrays of gate-defined quantum dots (QDs) are emerging as a promising candidate for storing spin qubits and serve as a platform for quantum computing applications.^{1–3)} Their promising potential arises from the ease of controlling key parameters,^{4–7)} the ability to rapidly measure spin and charge states,⁸⁾ and the long decoherence times they offer.^{9–11)} Previous advancements, including the demonstration of two-qubit gates and quantum algorithms,^{12–14)} have further strengthened their potential as a building block for solid-state quantum computers.

However, one of the primary challenges in using gate-defined QDs for large-scale quantum computing is the complex and time-consuming process of tuning these dots for QD array formation.^{15–18)} In this paper, tuning refers to the adjustment of the voltages applied to the electrostatic gates, which must be precisely controlled to trap individual electrons within the QDs. This process is crucial because each qubit must be in a well-defined quantum state to perform reliable quantum operations. Typically, the manual tuning process involves scanning the parameter space of gate voltages and observing the behavior of the QDs to identify the desired operating regime. However, this process is highly time-intensive, especially as the number of qubits in a system increases.

Several automated methods have been developed to tackle the challenges of tuning in quantum dot (QD) devices, aiming to simplify various steps in the tuning process. The choice of automation often depends on the device design, as each tuning step requires specific techniques. Some methods focus on the initial stages of tuning, transitioning the device from its starting configuration to a voltage regime where QDs can form.¹⁶⁾ Others concentrate on more complex tasks, such as configuring quantum dots—for instance, transitioning from two single quantum dots to coupled double quantum dots.¹⁹⁾ These methods aim to control the number of electrons within each QD or fine-tune the couplings in multi-QD systems.^{20–23)} They utilize a range of tools, including convolutional neural networks (CNNs),²⁴⁾ deep generative modeling,²⁵⁾ and classical feature extraction methods like the Hough transformation.²⁶⁾

Among the automated tuning methods developed for optimizing quantum dot devices, the Ray-Based Classification (RBC) framework offers a novel approach to reduce the measurement time and complexity involved in quantum dot tuning.²⁷⁾ Traditional tuning methods often rely on exhaustive two-dimensional (2D) voltage scans, where gate voltages are varied systematically across a grid. This process is time-consuming and inefficient because the number of measurements increases significantly with the resolution of the scan. While 2D scans provide comprehensive

information about the system, the RBC framework minimizes the number of measurements required by performing one-dimensional (1D) voltage sweeps along various directions, known as "rays." These rays exploit the freedom of angle in the multidimensional voltage space, allowing for efficient exploration with significantly fewer measurements. By strategically selecting the angles of these rays, the RBC framework balances the need for valuable system information with the practical constraints of time and resource efficiency.

By focusing on distinguishing between different charge states during quantum dot tuning, these rays provide qualitative information about the boundaries that separate various operational regimes of the quantum dots. This targeted approach enables a more efficient exploration of the parameter space and a faster identification of the desired charge configuration. In this way, the RBC framework complements existing methods by offering a more practical and time-effective means of achieving precise control over quantum dot charge states.

In this paper, operating within the RBC framework, we employ a Long Short-Term Memory (LSTM) neural network to create a closed-loop system for achieving single-dot formation in a GaAs/AlGaAs device.³²⁾ LSTM networks are well-suited for handling sequential data and have proven effective in identifying patterns in time-series data.²⁸⁾ By integrating the RBC framework's minimal one-dimensional sweeps with the LSTM neural network's pattern recognition capabilities, we develop a closed-loop system for automatically tuning separate and independent single quantum dots in a quadruple dot device.

Our autotuning algorithm operates with minimal prior knowledge, specifically the gate design parameters and the measured depletion value of a central gate shared by all quantum dots. We detail the algorithm and validate it by successfully tuning four distinct single quantum dots within a quadruple dot array. The algorithm autonomously identifies the appropriate gate voltages needed to tune all quantum dots into separate single-dot regimes, as confirmed by experimental results.